Vijay Baskar — AI Interview Bank

Full question bank structured Basic → Intermediate → Advanced. Red = question. Blue = simple answer + interview answer. Green = example from real work. 319 questions (77 basic, 115 intermediate, 127 advanced).

How to use this book. Intern / fresher: Level 1 end-to-end, then ML + HTTP/SQL from Level 2. Mid: Level 2, plus the 25 industry questions at the start of Level 3. Senior / staff / principal: Level 3, and be ready to drop to a Level 1 SQL index or Git force-push if they probe fundamentals. Speak only known metrics: $2M indoor-nav beacon savings, sub-100ms edge inference, sub-5% trading drawdown, 70% ANR reduction, 30% app performance, 500K users / 40% faster backends. TAN Intelligence is an independent venture, not a second employer.

Three levels

Level 1 — Basic (77 questions). Fresher fundamentals first: Python, OOP, data structures and Big-O, HTTP/REST/JSON/JWT, SQL, Git, Linux, Java/Kotlin/Android/Flutter, tests, CI, and config. If you cannot explain a LEFT JOIN or a decorator, do not skip ahead.

Level 2 — Intermediate (115 questions). Core ML, computer vision and spatial AI, FastAPI/Spring Boot backends, mobile production (ANR, coroutines, Flutter), LLM/RAG/agents intro, trading hygiene, and cloud. This is the mid-level IC loop: ship a model, an API, and a client without pretending to be a research lab.

Level 3 — Advanced (127 questions). The 25 most-asked industry questions first, then RAG at scale, agents/tools/memory, evals/safety/MLOps, serving (vLLM, KV cache, LoRA, SSE), system design, and leadership stories from ezyCart, indoor navigation, and TAN. Production GenAI, not a notebook.

Identity. This bank is for Vijay Baskar (LinkedIn vb1707). Do not mix with the Arjun Dev profile. Examples use Retailetics work (ezyCart, Lisa, vision/SLAM, ONNX/TF Lite) and Colan mobile/backend history. If you did not measure it, do not invent a dollar figure.

Level 1 — Basic (fresher fundamentals)

Python, OOP, data structures, HTTP, SQL, Git, Linux, Java/Kotlin/Flutter, tests and CI. Answer in two layers: a sentence a new grad would say, then the interview version.

Python

Q1. What is Python and why is it used in AI?

Answer (simple). Python is a readable language with lots of ready-made tools for data and models.

Answer (interview). Dynamic language with NumPy, Pandas, PyTorch, FastAPI. Fast to experiment; profile before optimizing. GIL matters for CPU threads; use multiprocessing or native extensions for heavy compute.

Example. AI services at Retailetics sit in Python FastAPI; training/export also in Python.

Example. A notebook prototype is not production until it has tests, types, and packaging.

Q2. What is the difference between a list, tuple, set, and dict?

Answer (simple). List is a line of items you can change. Tuple is a locked line. Set is a bag of unique stickers. Dict is a labeled locker.

Answer (interview). list: ordered, mutable, O(1) append. tuple: immutable, hashable if contents are. set: unique, O(1) avg lookup. dict: key→value hash map. Choose by mutability, uniqueness, and lookup needs.

Example. Cache SKUs in a set for O(1) membership. Cart lines are a list of dicts {sku, qty}.

Example. Dedup device event_ids with a set so the same scan is not charged twice.

Q3. What is a list comprehension?

Answer (simple). A short way to build a list from another list with a rule.

Answer (interview). [f(x) for x in xs if pred(x)] is eager. Generator ( ) is lazy. Nested comps hurt readability — prefer a loop if nested.

Example. skus = [line['sku'] for line in cart if line['qty'] > 0]

Q4. What are *args and **kwargs?

Answer (simple). Extra unnamed toys go in *args. Extra named toys go in **kwargs.

Answer (interview). *args is a tuple of extra positional args; **kwargs a dict of extra keywords. Useful for wrappers, decorators, and forwarding to FastAPI/Pydantic.

Example. A logging decorator wraps any tool function without rewriting its signature.

Q5. What is a decorator?

Answer (simple). A wrapper that adds behavior around a function without rewriting it.

Answer (interview). Syntactic sugar for f = deco(f). Common: @app.get, retries, auth, timing. Closures capture state. functools.wraps preserves metadata.

Example. Time a FastAPI inference handler with a decorator that logs p95-relevant duration.

Q6. What is a generator? What is yield?

Answer (simple). A generator gives you items one at a time instead of building the whole pile.

Answer (interview). yield suspends the frame. Generators are iterators, memory-light, good for streaming tokens or reading large files. Send/throw exist but keep it simple.

Example. Stream SSE tokens with a generator that yields each LLM chunk then stops on disconnect.

Q7. Mutable vs immutable types?

Answer (simple). Immutable means you cannot change the box; you make a new one. Mutable you can poke.

Answer (interview). Immutable: int, str, tuple, frozenset. Mutable: list, dict, set. Mutable default args are a classic bug (def f(x=[])). Hashable keys must be immutable.

Example. Never use {} as a default cache in a function — it is shared across calls.

Q8. What is the GIL?

Answer (simple). Python's lock so only one thread runs Python bytecode at a time.

Answer (interview). I/O-bound work can still use threads. CPU-bound Python should use multiprocessing, C extensions, or native inference (ONNX/TensorRT) off the GIL.

Example. TF Lite/ONNX inference releases into native code; the GIL is not the cart's bottleneck — the NPU/CPU is.

Q9. == vs is? shallow vs deep copy?

Answer (simple). == asks 'same value?'. is asks 'same object in memory?'. Shallow copy shares inner boxes; deep copy clones them.

Answer (interview). is compares identity. copy.copy vs copy.deepcopy. Nested dicts in agent state need deepcopy if you checkpoint.

Example. LangGraph state copies: mutating a nested list without copy can leak into the previous checkpoint.

Q10. What are exceptions and context managers?

Answer (simple). try/except is a safety net. with is a helper that always tidies up, like closing a file.

Answer (interview). Use specific exceptions. with open, with lock, contextlib.contextmanager. Never swallow Exception silently in payment/checkout paths.

Example. WebSocket send: catch disconnect, cancel the LLM job, do not retry a charge.

Q11. What is virtualenv / venv / pip / requirements vs lock files?

Answer (simple). A venv is a private toy box of libraries so projects do not fight.

Answer (interview). Pin versions (requirements.txt or poetry.lock/uv.lock). Reproducible Docker images. Never pip install unpinned in production.

Example. Cart service image pins FastAPI, Pydantic, onnxruntime — same versions in staging and prod.

Q12. What is typing in Python? Optional, list[str], Protocol?

Answer (simple). Type hints are labels that help humans and tools catch mistakes before runtime.

Answer (interview). PEP 484. Pydantic uses types at the API boundary. mypy/pyright in CI. Optional[X] = X | None. Prefer list[str] over List[str] in 3.9+.

Example. add_to_cart(sku: str, qty: int) — Pydantic rejects qty=-1 before the agent proceeds.

Q13. sync vs async Python? async/await?

Answer (simple). async is for waiting on the network without blocking the whole restaurant.

Answer (interview). asyncio event loop. FastAPI async def for I/O. Do not run CPU inference inside async without a threadpool — it stalls the loop.

Example. async WebSocket fan-in from devices; run ONNX infer in a worker pool.

Q14. What is a module vs package? __init__.py? PYTHONPATH?

Answer (simple). A module is one file. A package is a folder of modules.

Answer (interview). Imports, circular import pitfalls, explicit relative imports. Keep API surface small.

Example. Split retrieval, agents, and serving packages so tests can mock tools.

Object-oriented programming

Q15. What is OOP? Class vs object?

Answer (simple). A class is the cookie cutter. An object is a cookie.

Answer (interview). Encapsulation, abstraction, inheritance, polymorphism. Prefer composition over deep inheritance in services.

Example. A Detector base with YOLODetector and RTDETRDetector sharing a preprocess interface.

Q16. What are the four pillars of OOP?

Answer (simple). Hide guts (encapsulation), show a simple face (abstraction), reuse a parent (inheritance), same action different types (polymorphism).

Answer (interview). Interviewers want examples, not slogans. SOLID: SRP, OCP, LSP, ISP, DIP — apply lightly, not as religion.

Example. PaymentGateway interface with NFC and card implementations — same charge() call.

Q17. Overloading vs overriding? Abstract class vs interface?

Answer (simple). Overriding replaces a parent's method. Overloading is same name different args (Java). Python uses default args / *args instead.

Answer (interview). ABC/Protocol in Python; interface in Java/Kotlin. Abstract = cannot construct, must implement.

Example. Kotlin CheckoutUseCase overrides execute(); Java cannot override a static method as polymorphism.

Q18. What is composition vs inheritance?

Answer (simple). Inheritance is 'is-a'. Composition is 'has-a'. Prefer has-a when you just need a helper.

Answer (interview). Deep hierarchies break. Inject a Retriever into an Agent rather than Agent extends Retriever.

Example. Lisa has a retriever, a tool box, and a tracer — it is not a subclass of LlamaIndex.

Q19. What is encapsulation and why private fields?

Answer (simple). Keep the messy bits inside so callers cannot break invariants.

Answer (interview). Python _convension vs real privacy in Java/Kotlin. Validate in constructors.

Example. Cart total is computed from lines; callers should not set total=0 by hand.

Data structures and Big-O

Q20. What is Big-O? Best / average / worst case?

Answer (simple). A guess of how work grows when the pile of data grows.

Answer (interview). O(1), O(log n), O(n), O(n log n), O(n^2). State assumptions. Amortized vs worst. Memory also matters.

Example. Rerank 50 chunks with a cross-encoder is O(k) model calls — do not rerank 10M docs.

Q21. Array vs linked list vs hash map vs tree vs heap?

Answer (simple). Array is a row of seats. Linked list is a chain. Hash map is lockers. Tree is a family tree. Heap is a priority line.

Answer (interview). Arrays: cache-friendly. Hash maps: expected O(1). Trees/heaps: ordered / priority. Know when each fails (hash collisions, unbalanced trees).

Example. Event_id → processed boolean is a hash set. Next-soonest retry is a heap.

Q22. Stack vs queue vs deque? BFS vs DFS?

Answer (simple). Stack is last-in first-out (plates). Queue is first-in first-out (ticket line).

Answer (interview). DFS uses stack/recursion; BFS queue. LangGraph cycles need a visited/step budget like BFS/DFS on graphs.

Example. Agent loop detection is a visited set of (node, tool-signature).

Q23. What is a graph? Directed vs undirected?

Answer (simple). Dots (nodes) and arrows (edges). Directed arrows have a direction.

Answer (interview). LangGraph is a directed graph with state. DAGs have no cycles; agents often need bounded cycles.

Example. Planner → retriever → tool → answer, with a loop edge capped at N.

Q24. Binary search? When does it work?

Answer (simple). Guess the middle of a sorted list, throw away half, repeat.

Answer (interview). O(log n) on random-access sorted data. Off-by-one bugs are common. Interpolation search is a variant.

Example. Not for unsorted SKU lists — use a hash index instead.

Q25. Recursion vs iteration? What is a stack overflow?

Answer (simple). Recursion is a function calling itself. Too deep and the call stack explodes.

Answer (interview). Tail recursion is not optimized in Python. Prefer iteration for deep trees; explicit stack.

Example. Do not recurse unbounded on a tool-calling agent — use a step counter.

Q26. What is hashing? Collisions?

Answer (simple). A blender that turns a key into a locker number. Two keys can land in the same locker.

Answer (interview). Load factor, chaining vs open addressing, hashCode/equals contract in Java. Security: hash flooding.

Example. SKU strings hashed into a dict. Bad hashCode in Java breaks HashMap.

HTTP, REST, and JSON

Q27. What is HTTP? Methods GET POST PUT PATCH DELETE?

Answer (simple). HTTP is how browsers and apps talk to servers. GET reads. POST creates. PUT replaces. PATCH nips. DELETE removes.

Answer (interview). Safe/idempotent: GET should not change state. PUT/DELETE idempotent. POST often not. Status codes: 2xx success, 4xx client, 5xx server.

Example. GET /products/{sku}. POST /cart/items. Do not GET to charge a card.

Q28. What is REST?

Answer (simple). A style of API: resources with URLs, HTTP verbs, and stateless calls.

Answer (interview). Resource-oriented, HATEOAS rarely fully used. Versioning, pagination, error bodies. Contrast GraphQL (client picks fields) and RPC.

Example. Product catalogue REST; live cart telemetry is WebSocket, not REST polling.

Q29. What are HTTP status codes 200, 201, 204, 400, 401, 403, 404, 409, 429, 500, 503?

Answer (simple). 200 OK, 201 created, 400 you sent junk, 401 who are you, 403 you may not, 404 missing, 429 slow down, 500 we broke, 503 busy.

Answer (interview). 401 unauthenticated vs 403 unauthorized. 409 conflict/idempotency. 429 rate limit. 502/504 gateway. Do not return 200 with an error payload for tools.

Example. add_to_cart of an unknown SKU: 404. Duplicate event id: 200/204 with the original result (idempotent).

Q30. What is idempotency?

Answer (simple). Doing the same request twice should not double the side effect.

Answer (interview). Idempotency keys for payments and device events. PUT is idempotent; POST needs a key. Exactly-once is hard; at-least-once + idempotent handler is the usual design.

Example. Scan event_id already processed → ignore. Payment key reused → return original receipt.

Q31. Headers, cookies, CORS, TLS?

Answer (simple). Headers are sticky notes on the request. TLS is the encrypted tunnel. CORS is the browser's guest list.

Answer (interview). Authorization: Bearer. Content-Type. CORS is a browser rule, not API security. TLS everywhere. Cookies: HttpOnly, Secure, SameSite.

Example. Mobile app uses bearer tokens; do not put secrets in query strings.

Q32. What is JSON? JSON vs XML vs protobuf?

Answer (simple). JSON is nested labeled boxes of text, numbers, lists. XML is older tagged text. Protobuf is compact binary.

Answer (interview). JSON for public APIs; protobuf/Avro for high-QPS internal events. Schema evolution matters.

Example. LLM tool calls are JSON. IoT high-rate telemetry might be protobuf on the wire.

Q33. Query params vs path vs body?

Answer (simple). Path is which thing. Query is filters. Body is the payload.

Answer (interview). Do not put PII in query logs. GET has no body in practice. Pagination: cursor > offset for large catalogues.

Example. GET /products?q=pasta. POST /assist with the basket JSON in the body.

Q34. REST vs GraphQL vs gRPC vs WebSockets vs SSE?

Answer (simple). REST is ask/answer. GraphQL is pick-your-fields. gRPC is fast binary RPC. WebSockets are a phone call. SSE is a one-way radio.

Answer (interview). Choose by chat vs events vs CRUD. LLM tokens: SSE. Device telemetry: WebSocket. Catalogue: REST/GraphQL.

Example. Lisa streams SSE; trolley sensors use WebSockets into FastAPI/Spring Boot.

SQL

Q35. What is SQL? Primary key vs foreign key?

Answer (simple). SQL talks to tables. A primary key uniquely names a row. A foreign key points at another table's key.

Answer (interview). Normalization vs denormalization. Constraints enforce integrity that app code will forget.

Example. orders.id PK. orders.customer_id FK. Do not orphan cart lines.

Q36. INNER vs LEFT vs RIGHT vs FULL join?

Answer (simple). INNER is only matches. LEFT keeps all left rows even without a match.

Answer (interview). NULLs in outer joins. Accidental INNER drop of unmatched products is a common bug in analytics.

Example. LEFT JOIN products to inventory so missing stock still shows the SKU.

Q37. WHERE vs HAVING? GROUP BY?

Answer (simple). WHERE filters rows before grouping. HAVING filters after aggregates.

Answer (interview). SELECT sku, SUM(qty) FROM lines GROUP BY sku HAVING SUM(qty)>10. Index columns used in WHERE.

Example. Top SKUs sold per day for a merchant dashboard.

Q38. What is an index? Clustered vs secondary? When does it hurt?

Answer (simple). An index is the book's back index — faster find, slower writes, extra storage.

Answer (interview). B-tree vs hash vs GIN (pgvector). Composite order matters. Use EXPLAIN ANALYZE. Too many indexes slow checkouts.

Example. INDEX on orders(customer_id, created_at). EXPLAIN before claiming a 40% API speedup.

Q39. What is a transaction? ACID?

Answer (simple). A transaction is a bundle that all happens or none happens. ACID: Atomic, Consistent, Isolated, Durable.
Answer (interview). Isolation levels: read committed vs repeatable read vs serializable. Deadlocks. Saga for distributed checkout.
Example. Charge + emit receipt in one transaction; do not charge if insert fails.

Q40. Normalization (1NF, 2NF, 3NF) vs denormalization?

Answer (simple). Normalization reduces duplicate facts. Denormalization copies facts for speed.
Answer (interview). OLTP vs OLAP. Event sourcing as an alternative. Know update anomalies.
Example. Catalogue name denormalized onto a receipt line so historical receipts do not change when the SKU is renamed.

Q41. SQL injection and how to prevent it?

Answer (simple). Never glue user text into SQL like tape. Use parameters.
Answer (interview). Prepared statements, ORMs, least-privilege DB users. MCP/SQL tools are dangerous without allowlists.
Example. sku = request; execute('... WHERE sku=%s', [sku]) — never f-strings.

Q42. NULL behavior in SQL? Three-valued logic?

Answer (simple). NULL means unknown. NULL = NULL is not true.
Answer (interview). IS NULL. COALESCE. Aggregates skip NULLs. Unique indexes and multiple NULLs depend on the engine.
Example. missing gluten_free flag is NULL, not false — Lisa must not claim 'not gluten-free'.

Q43. OFFSET pagination vs keyset/cursor?

Answer (simple). OFFSET skips rows (gets slow). Cursor remembers the last id and continues.
Answer (interview). Large catalogues: keyset on (created_at, id). OFFSET 100000 is a full scan.
Example. Product admin list uses cursor pagination, not page=5000.

Q44. What is N+1 query problem?

Answer (simple). You fetch a list, then one extra query per item. That is a thousand tiny trips.
Answer (interview). Eager load/join or IN (...) batch. ORMs hide this. Trace SQL in staging.
Example. Loading 50 cart lines then 50 product queries — join or batch instead.

Git

Q45. What is Git? commit, branch, merge, rebase?

Answer (simple). Git is a time machine for code. A commit is a snapshot. A branch is an alternate timeline.
Answer (interview). Merge creates a merge commit; rebase replays. Never rebase public shared history casually. PR reviews.
Example. Feature branch cursor/interview-guide-a8c9 merged to main, then gh-pages publish.

Q46. git clone, status, add, commit, push, pull, fetch?

Answer (simple). clone copies. status looks. add stages. commit saves. push shares. pull fetches+merges.
Answer (interview). fetch vs pull. Origin vs upstream. Detached HEAD. .gitignore.
Example. Do not commit .env or model secrets. Pin PDFs that are meant to be public.

Q47. merge conflict? how to resolve?

Answer (simple). Two people edited the same lines. You pick the combined truth.
Answer (interview). Open the markers, test, commit. Prefer smaller PRs to reduce conflicts.
Example. Two prompt files edited — keep both eval cases, do not drop tests.

Q48. What is a pull request / code review?

Answer (simple). A request for teammates to read your change before it lands.
Answer (interview). Describe why, risk, test plan. Review for correctness, security, not nits only.
Example. Prompt change PRs must attach eval before/after scores.

Q49. HEAD, stash, tag, cherry-pick?

Answer (simple). HEAD is where you are. stash pockets dirty files. tag names a version. cherry-pick copies one commit.
Answer (interview). Tags for releases (model-prompt-v12). Cherry-pick hotfixes carefully.
Example. Tag last-known-good prompt for rollback.

Q50. force push risks?

Answer (simple). force push rewrites shared history and can erase teammates' work.
Answer (interview). Use --force-with-lease if you must. Protect main.
Example. Never force-push main/gh-pages of a public site without checking.

Linux, OS, and networking

Q51. What is a process vs thread?

Answer (simple). A process is a program. Threads are workers inside it sharing memory.
Answer (interview). Isolation vs shared-memory races. GIL. Containers isolate processes.
Example. One FastAPI process, threadpool for ONNX; separate process for vLLM.

Q52. stdin, stdout, stderr, exit codes?

Answer (simple). stdin in, stdout out, stderr complaints. Exit 0 means OK.
Answer (interview). CI fails on non-zero. Log structured JSON to stdout in K8s.
Example. Eval job exits 1 if groundedness drops — that blocks deploy.

Q53. What is DNS, IP, port, TCP vs UDP?

Answer (simple). IP is the building address. Port is the apartment. TCP is a reliable call. UDP is a postcard.
Answer (interview). TLS on TCP. WebSockets upgrade HTTP. UDP for some realtime but not checkout.
Example. Cart app connects wss:// to the IoT gateway on 443.

Q54. What is a deadlock? Race condition?

Answer (simple). Deadlock: two waiters each hold what the other needs. Race: winner depends on luck of timing.
Answer (interview). Locks, lock ordering, immutable messages, actor/queue models. Reproduce with stress tests.
Example. Two services updating the same cart line without optimistic versioning.

Java, Kotlin, Android, and Flutter (fresher)

Q55. Java vs Kotlin for Android?

Answer (simple). Kotlin is the modern Android language: less boilerplate, null-safety.
Answer (interview). Null safety, coroutines, data classes, interop with Java. Jetpack Compose is Kotlin-first. Java still appears in older modules.
Example. ezyCart Android app: Kotlin + existing Java libraries.

Q56. What is null safety in Kotlin?

Answer (simple). String? might be missing. String cannot. The compiler nags you to check.
Answer (interview). ?. ?: !! (avoid !!). Java interop platform types. Prevents a class of NPEs.
Example. sku: String not String? on add_to_cart.

Q57. What is an Activity / Fragment / ViewModel?

Answer (simple). Activity is a screen. Fragment a piece of screen. ViewModel keeps data when the screen rotates.
Answer (interview). MVVM: UI observes ViewModel; ViewModel talks to repositories. SavedStateHandle.
Example. Cart screen ViewModel holds line items; rotation does not empty the cart.

Q58. What is Jetpack Compose vs XML views?

Answer (simple). Compose describes UI as functions of state. XML was nested layout files.
Answer (interview). Recomposition, remember, unidirectional data flow. Interop with views exists.
Example. Staff Flutter app is another declarative UI; Android cart may use Compose.

Q59. What is Flutter / Dart? Widget tree?

Answer (simple). Flutter paints its own UI on iOS and Android from one Dart codebase.
Answer (interview). Immutable widgets, setState/BLoC/Provider, platform channels. Good for staff tools; native may still win for tight hardware.
Example. Flutter staff/TV monitor and Flutter SDK lite for small vendors.

Q60. What is an APK / Play release / ProGuard?

Answer (simple). APK is the install package. Play is the store. ProGuard/R8 shrinks and obfuscates.
Answer (interview). Signing, versionCode, staged rollout, mapping files for crash deobfuscation. CI with Jenkins/GitLab/Bitrise.
Example. Colan CI/CD cut ANR 70% and lifted performance 30% — know how you measured ANR.

Q61. HTTP on mobile: Retrofit/OkHttp vs Ktor?

Answer (simple). Libraries that call REST from the phone.
Answer (interview). Interceptors for auth, retries, logging (no PII). Timeouts. Certificate pinning when needed.
Example. Android ezyCart calls backend; WebSocket client for live IoT.

Q62. What is MVVM vs MVP vs MVC?

Answer (simple). Ways to split UI from logic so screens do not become god objects.
Answer (interview). MVVM + LiveData/Flow on Android; BLoC on Flutter. Test ViewModels without the UI.
Example. CartViewModel unit tests do not need an emulator.

Q63. Coroutines / Flow vs RxJava vs callbacks?

Answer (simple). Coroutines are Kotlin's readable async. Flow is a stream of values.
Answer (interview). structured concurrency, cancellation, Dispatchers.IO vs Main. Prefer structured concurrency over callback hell.
Example. Collect device events on IO, update UI on Main.

Q64. What is SOLID in one sentence each?

Answer (simple). Single job, open to extension, don't break subclasses, small interfaces, depend on abstractions.
Answer (interview). Use as a smell detector, not a checklist to gold-plate a trolley app.
Example. A 2000-line Activity violates SRP — split scanning, pay, and ads.

Tests, CI, security, and config

Q65. What is JSON Schema / OpenAPI?

Answer (simple). A contract that says what fields exist and their types. OpenAPI documents REST APIs.
Answer (interview). FastAPI generates OpenAPI from Pydantic. Breaks are versioned. Clients stub from the spec.
Example. Tool JSON schemas for agents are the same idea as OpenAPI for humans.

Q66. What is CI vs CD?

Answer (simple). CI is auto-test on every change. CD is auto-release when tests pass.
Answer (interview). Unit, lint, security scan, evals, image build, staging, prod with rollback.
Example. Prompt change blocked if eval groundedness drops.

Q67. Unit vs integration vs e2e tests?

Answer (simple). Unit is one function. Integration is several pieces. e2e is the whole user journey.
Answer (interview). Test pyramid. Mock LLMs in unit tests; contract-test tools; a few e2e cart flows.
Example. Unit-test Pydantic tool args. Integration-test retriever+rerank with a fixture index.

Q68. What is a race in checkout if two scans arrive?

Answer (simple). Two events at once can double-count if you are not careful.
Answer (interview). Per-cart lock or serial queue + idempotent event ids. Optimistic versioning on cart row.
Example. Same barcode twice: qty+=1 once per unique event_id.

Q69. How do you read a stack trace?

Answer (simple). Start at the bottom cause, then the frames that led there.
Answer (interview). Caused by vs traceback. Map obfuscated Android stacks with mapping.txt.
Example. ANR traces point at main-thread IO — move work off main.

Q70. What is UTF-8?

Answer (simple). A way to store every language's characters as bytes.
Answer (interview). Do not confuse character length with byte length. JSON default UTF-8.
Example. Tamil recipe names must round-trip in the assistant.

Q71. Environment variables vs config files vs secret managers?

Answer (simple). Config is settings. Secrets are passwords. Do not commit secrets.

Answer (interview). 12-factor. Kubernetes secrets / vault. Rotate. Different env per stage.

Example. Bedrock keys never in the Android app; they live on the server.

Q72. What is pagination, filtering, sorting in APIs?

Answer (simple). Do not send the whole catalogue. Send a page, with filters and order.

Answer (interview). Stable sort keys, max page size, total vs cursor. Reject unbounded queries.

Example. Staff stock-audit lists aisles page by page.

Q73. What is a 12-factor app (name a few)?

Answer (simple). A set of habits: config in env, stateless processes, logs as streams, etc.

Answer (interview). You need not recite all 12; show you build scalable services.

Example. Cart API is stateless; session in Redis/DB; logs to stdout.

Q74. GET caching and Cache-Control / ETag?

Answer (simple). Browsers and CDNs can reuse a GET if you allow it.

Answer (interview). ETag/If-None-Match. Authorization headers often bypass shared caches. Do not cache personalized prices.

Example. Public product images cached; membership price never.

Q75. What is a JWT? Risks?

Answer (simple). A signed blob saying who you are. Anyone with the blob looks like you until it expires.

Answer (interview). exp, iss, aud. Store securely on mobile. Revocation is hard — short TTL + refresh. Never put in localStorage on web if XSS.

Example. Staff app JWT cannot call shopper-only routes (403).

Q76. Hash vs encryption vs encoding?

Answer (simple). Hash is one-way blender. Encryption is a lock with a key. Encoding is changing shape (Base64), not secret.

Answer (interview). Passwords: salted hash (bcrypt/argon2). PII at rest: encrypt. Base64 is not encryption.

Example. Do not Base64 a PAN and call it 'encrypted' in an LLM prompt.

Q77. What is stdout logging vs a debugger?

Answer (simple). Logs tell the story later. A debugger is live stepping.

Answer (interview). Structured logs with request_id. Debug locally; traces in prod (LangSmith for LLM).

Example. Correlate cart_id across Android, WebSocket, and RAG spans.

Level 2 — Intermediate (core ML, CV, backend, mobile)

Machine learning, computer vision, APIs, mobile production, LLM/RAG intro, trading, cloud. Connect every answer to a system you would actually ship.

Machine learning fundamentals

Q78. Supervised vs unsupervised vs reinforcement learning?

Answer (simple). Supervised: teacher gives answers. Unsupervised: group similar things. RL: score from trial and error.

Answer (interview). Supervised needs labels (detection boxes, sentiment). Unsupervised: clustering, embeddings. RL: agents with rewards — expensive and unstable for checkout; prefer supervised + rules + evals.

Example. YOLO is supervised detection. Recipe retrieval uses embeddings (unsupervised pretrain + supervised eval). Do not RL a payment agent in production first.

Q79. Train / validation / test split? Why not peek at test?

Answer (simple). Practice, dress rehearsal, real exam. If you keep retaking the exam while studying, the score lies.

Answer (interview). Fit only on train. Tune on val. Report test once. Time-series: chronological splits. Nested CV when tuning is heavy.

Example. Trading labels shuffled randomly leak the future. Vision: hold out a store not seen in training.

Q80. Overfitting vs underfitting? Regularization?

Answer (simple). Overfit memorizes homework. Underfit is too simple. Regularization is a 'don't be fancy' penalty.

Answer (interview). L2/L1, dropout, early stop, data aug, smaller models. Learning curves. Bias-variance talk without cargo-cult.

Example. A detector with 99% train mAP and 40% on night-shift cameras is overfit to daytime.

Q81. Bias vs variance?

Answer (simple). Bias is always missing the dartboard the same way. Variance is scattered shots.

Answer (interview). High bias: add features/model capacity. High variance: more data, regularization. Error decomposition is a diagnostic, not a slogan.

Example. A linear model on SKU images underfits; a huge net on 200 labels overfits.

Q82. Precision, recall, F1, accuracy, ROC-AUC, PR-AUC?

Answer (simple). Accuracy can lie if 99% of frames have no product. Precision: of alarms, how many true. Recall: of truths, how many found.

Answer (interview). Imbalanced detection: PR curve, F-beta. Thresholds from cost of false charge vs missed item. Confusion matrix first.

Example. False-positive SKU at checkout is a precision problem (shopper charged wrong). Missed item is recall.

Q83. What is a loss function? Cross-entropy vs MSE?

Answer (simple). Loss is a score of how wrong the model is. Training tries to shrink it.

Answer (interview). CE for classification, MSE/MAE for regression, specialized detection losses (box + class + objectness). Loss ≠ business metric.

Example. Optimize detection loss, but ship on checkout-error rate, not only mAP.

Q84. Gradient descent, learning rate, batch size?

Answer (simple). Take steps downhill. Learning rate is step size. Batch is how many examples per step.

Answer (interview). SGD/Adam. Too large LR diverges; too small crawls. Large batch needs LR scaling. Mixed precision.

Example. Fine-tune LoRA with a small LR on the adapter; do not blow up the frozen base.

Q85. Backpropagation in one minute?

Answer (simple). Send the error backward so each layer knows how to change.

Answer (interview). Chain rule through the graph. Autograd in PyTorch. Vanishing/exploding gradients; residuals, init, clipping.

Example. You do not hand-derive YOLO backprop; you must know what a NaN loss means.

Q86. What is a neural network? MLP vs CNN vs transformer?

Answer (simple). MLP is stacked fully-connected layers. CNN looks at local image patches. Transformer uses attention.

Answer (interview). Inductive bias: CNNs for grid locality; transformers for long-range. Modern detectors mix both (RT-DETR).

Example. Cart camera: CNN/transformer detector. Text RAG: transformer LLM + embedding model.

Q87. Embeddings for non-text (images)?

Answer (simple). Turn a photo into a point on a map so similar photos sit near each other.
Answer (interview). Contrastive (CLIP), metric learning. Use for visual search; still need SKU classifier/detector for checkout.
Example. Find similar products from a shelf photo; do not bill from embedding nearest-neighbor alone without a threshold.

Q88. Data leakage?

Answer (simple). Letting the exam answers into homework.
Answer (interview). Target in features, future timestamps, duplicate customers across splits, scaler fit on all data. Audit features.
Example. Using tomorrow's close in today's trading feature. Fitting StandardScaler on train+test.

Q89. Cross-validation vs walk-forward?

Answer (simple). CV rotates folds. Walk-forward slides a window through time.
Answer (interview). Random K-fold leaks time. Trading and demand: walk-forward, embargo, purge.
Example. TAN indicators: walk-forward, not 5-fold shuffle.

Q90. Class imbalance? How to handle?

Answer (simple). If 1 in 100 frames has a rare SKU, the model can ignore it and still look accurate.
Answer (interview). Resample, class weights, focal loss, threshold tuning, collect more rare data. Report per-class recall.
Example. Rare SKUs at checkout need extra labels, not just more bananas.

Q91. Feature scaling? Normalization vs standardization?

Answer (simple). Put features on a similar yardstick so one does not shout over others.
Answer (interview). Fit scaler on train only. Trees (XGBoost) care less; nets care more. Quantiles for heavy tails.
Example. Tabular promo model: scale numeric features; leave one-hot as 0/1.

Q92. XGBoost / gradient boosting vs random forest vs linear model?

Answer (simple). Linear is a straight rule. Forest is many trees voting. Boosting is trees that correct the last tree's mistakes.
Answer (interview). Tabular default: linear baseline then XGBoost. Need interactions and non-linearities. Neural nets when you have huge data or mixed modalities.
Example. Who buys which promo: XGBoost baseline before an LLM classifier.

Q93. What is sklearn Pipeline and why?

Answer (simple). A recipe that packages preprocess + model so you cannot forget a step.
Answer (interview). Preprocessing inside the pipeline, fit on train folds only. Joblib dump the whole pipeline.
Example. imputer → scaler → XGBoost in one object for merchant forecasting.

Q94. Calibration of probabilities?

Answer (simple). A 70% score should be right about 70% of the time.
Answer (interview). Reliability diagrams, Platt/isotonic. Detection confidence is not always a probability.
Example. Do not treat YOLO 0.9 as 90% chance without calibration on store data.

Q95. What is batch vs online vs mini-batch learning?

Answer (simple). Batch sees all data. Online sees one at a time. Mini-batch is the usual middle.
Answer (interview). Streaming drift: periodic retrain vs true online. Most production CV is mini-batch train, batch/offline eval, online infer.
Example. Nightly retrain on new labelled frames; inference per camera frame.

Q96. Train/serving skew?

Answer (simple). The model trained on clean tables but production feeds it messy live data.
Answer (interview). Same feature code path. Log features at serving for replay. Schema checks.
Example. Training used RGB 640px; device sent NV12 320px — mAP collapses. Match preprocess.

Q97. What is a baseline and why start there?

Answer (simple). A dumb simple method you must beat.
Answer (interview). Majority class, last-value forecast, BM25-only RAG, keyword FAQ. If you cannot beat it, do not ship complexity.
Example. BM25-only shopping FAQ vs full RAG — measure groundedness lift.

Q98. PyTorch: tensor, autograd, nn.Module, DataLoader?

Answer (simple). Tensor is a number grid. Autograd tracks how to change it. Module is a model. DataLoader feeds batches.
Answer (interview). device (cpu/cuda), no_grad for infer, eval() vs train() for dropout/BN. AMP.
Example. Export YOLO to ONNX with a dummy tensor of the production size.

Q99. eval() vs no_grad() vs inference_mode()?

Answer (simple). eval changes layers like dropout. no_grad stops tracking gradients. Both needed at serve time.
Answer (interview). Forgetting eval() leaves dropout on — noisy checkout detections.
Example. ONNX Runtime session is inherently infer; PyTorch proto must set eval.

Q100. ONNX export pitfalls?

Answer (simple). Not every PyTorch op exists in ONNX. Dynamic shapes and NMS are usual pain.
Answer (interview). Opset version, simplify, test numerical parity, then TensorRT/TF Lite. Keep a golden image test.
Example. A custom NMS plugin broke TensorRT; we compared boxes on a fixture clip.

Q101. JAX: grad, jit, vmap in one line each?

Answer (simple). grad = derivatives. jit = compile. vmap = run many in parallel.
Answer (interview). Functional, XLA. Different from PyTorch's imperative style. Choose by team and ecosystem.
Example. If the team is PyTorch-native for CV, do not rewrite YOLO in JAX for fashion.

Q102. TensorFlow vs PyTorch vs JAX — how do you choose?

Answer (simple). Three workshops. Pick the one with models, deploy path, and people.
Answer (interview). TF Lite path vs ONNX vs JAX/XLA. Reversible choice: export ONNX when possible.
Example. Detection in PyTorch → ONNX → TF Lite on Android.

Q103. What is transfer learning?

Answer (simple). Start from a model that already saw many pictures/words, then teach your small dataset.
Answer (interview). Freeze backbone, train head, then unfreeze lightly. Domain gap: supermarket ≠ COCO.
Example. Fine-tune YOLO on store SKUs rather than train from scratch.

Q104. What is data augmentation for images? Risks?

Answer (simple). Fake extra pictures by flipping, cropping, lighting changes.
Answer (interview). Augment only train. Photometric vs geometric. Do not flip text/OCR. Keep real night-aisle data too.
Example. Do not color-jitter so hard that SKU colors become unrecognizable.

Q105. Label noise and how it shows up?

Answer (simple). Wrong boxes teach the model wrong habits.
Answer (interview). Consensus labeling, QA sampling, loss robust to noise. Per-class confusion.
Example. Misboxed 'oil' as 'sauce' will cause checkout disputes — fix labels first.

Q106. mAP, IoU, NMS for detection?

Answer (simple). IoU is overlap of two boxes. NMS throws away extra boxes on the same object. mAP summarizes precision across classes and thresholds.
Answer (interview). IoU 0.5 vs 0.75 protocols. Soft-NMS. RT-DETR aims to avoid classic NMS. High mAP can still fail on glare/occlusion.
Example. Two boxes on one bottle: NMS keeps one. Report mAP@0.5 and checkout-oriented metrics.

Q107. Classification vs detection vs segmentation vs tracking vs OCR?

Answer (simple). Class: what. Detect: what+where box. Segment: pixels. Track: same object over time. OCR: read text.
Answer (interview). Checkout may need detect+track so a banana is not counted every frame.
Example. OCR for shelf labels; detection for items in the cart camera.

Computer vision and spatial AI

Q108. What is computer vision in production?

Answer (simple). Cameras plus models plus hardware plus monitoring — not a notebook demo.
Answer (interview). Define task, metric, lighting/occlusion domain, latency, privacy, fallback. Dataset flywheel.
Example. ezyCart self-checkout vision plus kiosk. Fail closed on low confidence.

Q109. YOLO family (v8/v10/v11) — what to say?

Answer (simple). A family of fast one-shot detectors. Bigger versions are more accurate and slower.

Answer (interview). Choose size by device budget. License, export path, NMS. Benchmark your video, not COCO blog posts.

Example. Cart NPU: nano/small. Back-office GPU: larger. Same eval set.

Q110. What is RT-DETR?

Answer (simple). A real-time detection transformer. Different guts than YOLO; still a detector.

Answer (interview). Set prediction, typically no NMS in the core. Compare mAP, small objects, ms, memory on the same device.

Example. Bake-off YOLO vs RT-DETR on aisle clips with identical image size and precision.

Q111. OpenCV's role if you have YOLO?

Answer (simple). OpenCV is the Swiss Army knife: resize, undistort, draw, classic vision.

Answer (interview). Pre/post around the net. Camera calibration. Do not rewrite undistort in PyTorch on device.

Example. Undistort cart camera, then run ONNX.

Q112. Camera calibration, distortion, extrinsics?

Answer (simple). Lenses bend pictures. Calibration measures the bend so geometry is honest.

Answer (interview). Intrinsics, extrinsics, IMU-camera extrinsics for VIO. Bad calibration → SLAM drift.

Example. Kimera-VIO needs camera-IMU calibration and time sync.

Q113. What is SLAM?

Answer (simple). Build a map while you figure out where you are.

Answer (interview). Front-end features, back-end optimization, loop closure. Failure: textureless walls, moving people, lighting.

Example. Indoor nav replaced thousands of Bluetooth beacons, saving over $2M — still explain aisle-level error and failure modes.

Q114. VIO vs VO vs wheel odometry?

Answer (simple). VO is camera-only motion. VIO adds IMU. Wheels slip.

Answer (interview). Complementary sensors. Sync clocks. Bias estimation.

Example. Trolley on smooth tiles: IMU helps during blur; wheels may slip — VIO is the fusion.

Q115. Kimera-VIO — what is it and what must you not overclaim?

Answer (simple). Open-source visual-inertial odometry in the Kimera family.

Answer (interview). Know inputs (mono/stereo + IMU), outputs (pose), assumptions, compute. Validate drift on store routes. Do not claim 'SLAM solved' if you only run VIO.

Example. Use it as the motion estimate feeding the product-finder path.

Q116. Spatial AI?

Answer (simple). Not just 'this is a chair' but 'the chair is 2 meters left'.

Answer (interview). Perception + geometry + localization + maybe planning. Coordinate frames (camera, IMU, map, store).

Example. Product finder: SKU in map frame, trolley in map frame, path between them.

Q117. Edge vs cloud inference for vision?

Answer (simple). Edge is on the device. Cloud is far away. Edge is faster and works if Wi-Fi dies; cloud is heavier models.

Answer (interview). Privacy, bandwidth, thermal, updates, fallback. Hybrid: tiny on-device + cloud for hard cases.

Example. Sub-100ms ONNX/TF Lite on cart; do not stream every frame to Bedrock.

Q118. TensorRT vs TF Lite vs ONNX Runtime?

Answer (simple). ONNX is the suitcase. TensorRT is a fast NVIDIA unpacker. TF Lite is the phone/edge unpacker.

Answer (interview). Operator coverage, quantization tools, profiling. Golden tests after every export.

Example. Android cart: TF Lite/ORT. NVIDIA kiosk: TensorRT.

Q119. How do you get sub-100ms vision?

Answer (simple). Time every stage. Shrink the slowest. Recheck quality.

Answer (interview). Resolution, quantization, batch=1 vs batch, CPU affinity, memcpy. End-to-end vs model-only — know which the resume number is.

Example. Resize was 40ms, infer 30ms, NMS 5ms — fix resize first.

Q120. Tracking IDs across frames?

Answer (simple). Give the same banana the same name as it moves so you do not count it 30 times.

Answer (interview). ByteTrack/SORT-style, IoU + appearance. Handle occlusion. Reset on cart-empty.

Example. Scan-bag-pay: track objects on the tray, not per-frame independent detections.

Q121. Lighting, glare, occlusion — domain shift?

Answer (simple). A model trained at noon fails at night or under a reflection.

Answer (interview). Per-scenario eval. Targeted data. Auto-exposure. Confidence gating.

Example. Store LED vs window glare sets — split metrics, not one mAP.

Q122. Privacy for cameras in a store?

Answer (simple). Do not keep faces if you do not need them. Say what you store.

Answer (interview). Retention, blur, on-device processing, access logs, regulation. Minimize.

Example. Checkout vision can drop frames that are not needed for the SKU event.

Backend, APIs, and DevOps

Q123. Why FastAPI for AI services?

Answer (simple). Python API with type checks and auto docs, sitting next to ML libraries.

Answer (interview). Pydantic validation, async, OpenAPI, deps. Offload CPU infer. Timeouts, auth, tracing.

Example. Vision and RAG orchestration in FastAPI; Java estate in Spring Boot.

Q124. Pydantic / bean validation?

Answer (simple). Check the form before you accept it.

Answer (interview). Boundary validation. Useful errors. Version schemas.

Example. LLM JSON and HTTP bodies both go through Pydantic — untrusted input.

Q125. Spring Boot in one minute?

Answer (simple). Java way to build services with batteries: web, security, config, actuators.

Answer (interview). Coexist with FastAPI behind stable contracts. Observability, thread pools, backpressure.

Example. IoT WebSocket gateway in Spring Boot talking to the Android app.

Q126. Microservices vs modular monolith?

Answer (simple). Many small services vs one well-cut program. Split only when the boundary pays rent.

Answer (interview). Network failure, distributed data, versioning. Start modular; extract when scaling or ownership demands.

Example. Do not split 'add to cart' into 12 services. Do split vision infer if it scales differently.

Q127. How do WebSockets stay reliable for IoT?

Answer (simple). Keep the call up, number the messages, retry without doubling, notice death.

Answer (interview). Device identity, schema, timestamps/seq, heartbeats, backoff, ack, idempotency, replay. Restore state on reconnect.

Example. ezyCart device ↔ Android via FastAPI/Spring Boot sockets — duplicate scans must not double charge.

Q128. REST vs WebSocket vs SSE in the trolley stack?

Answer (simple). REST for things. Sockets for live device chatter. SSE for token streams to the UI.

Answer (interview). Do not multiplex everything on one socket without a reason. Backpressure differs.

Example. SKU lookup REST; IMU/scan events WS; Lisa tokens SSE.

Q129. How do you debug a slow API?

Answer (simple). Measure first. Find the slow station. Change one thing. Measure again.

Answer (interview). p50/p95/p99, traces, EXPLAIN, N+1, payload size, downstream. Indexes vs cache.

Example. Colan 40% faster responses: know if it was indexing, caching, or query rewrite.

Q130. Cache invalidation basics?

Answer (simple). A sticky note of the answer. If the truth changes, throw the note away.

Answer (interview). TTL, explicit purge on catalogue update, stampede, stampede locking. Never cache ACL-sensitive personalized prices without a key.

Example. Recipe RAG semantic cache keyed by tenant + basket hash.

Q131. Message queues vs request/response?

Answer (simple). A queue is a waiting line for work that can happen in a second, not instantly.
Answer (interview). Celery/SQS/Kafka. At-least-once. Idempotent consumers. Backpressure.
Example. Embed 10k new products in Celery; HTTP request only enqueues.

Q132. Celery / background jobs?

Answer (simple). Do slow homework in the back room so the counter stays fast.
Answer (interview). Retries, acks, poison queues, idempotency, visibility timeout.
Example. Nightly re-embed of changed catalogue pages.

Q133. GraphQL when, not when?

Answer (simple). Client asks for exactly the fields it wants. Easy to over-fetch on the server instead.
Answer (interview). N+1, query cost analysis, persisted queries. Not a replacement for events.
Example. Staff app product details via GraphQL; not live IMU.

Q134. Authn vs authz? RBAC vs ABAC?

Answer (simple). Authn: who are you. Authz: what may you do. RBAC is roles. ABAC is attributes/policies.
Answer (interview). Per-resource checks. Tenant isolation. Service identity (mTLS).
Example. Shopper cannot read other carts. Staff role cannot export PII dumps without audit.

Q135. Rate limiting and backpressure?

Answer (simple). Slow down noisy clients so one shopper cannot knock over the API.
Answer (interview). Token bucket, 429, retry-after, load shedding, queue limits. Fairness per tenant.
Example. LLM gateway: tokens/min per store so one bot cannot burn the budget.

Q136. Health checks vs readiness vs liveness?

Answer (simple). Liveness: restart me if I am dead. Readiness: do not send traffic yet.
Answer (interview). Do not make liveness call the GPU if that hangs — deadlock restarts.
Example. ORT session ready = ready; crash loop ≠ ready.

Q137. Docker: image, layer, registry, non-root?

Answer (simple). A box with the app and its libraries. Layers cache builds. Registry stores boxes.
Answer (interview). Small base, pin digests, scan CVEs, non-root, no secrets in layers, HEALTHCHECK.
Example. FastAPI+ORT image, multi-stage build, distroless/runtime slim.

Q138. docker vs VM vs process?

Answer (simple). VM virtualizes hardware. Container virtualizes the OS userland. Process is just a program.
Answer (interview). Isolation vs overhead. Nested not needed on device.
Example. Cloud API in containers; on-cart infer is a process/library.

Q139. CI/CD for mobile + backend together?

Answer (simple). Two pipelines that meet at a contract test.
Answer (interview). Android Bitrise/GitLab + API tests. Version the API. Feature flags.
Example. App expecting new event schema must not ship before the gateway.

Q140. Observability: logs, metrics, traces?

Answer (simple). Logs are stories. Metrics are numbers. Traces are the path of one request.
Answer (interview). RED/USE. Exemplars. Cardinality explosions. Trace LLM + SQL + WS in one id.
Example. cart_id / trace_id from Android through RAG.

Q141. What is p95 latency and why not only average?

Answer (simple). Average hides the slow shoppers. p95 is the wait that 95% beat.
Answer (interview). SLOs on p95/p99. Tail from GC, JIT, GC, cold start, GC, GC — and retrieval.
Example. Lisa TTFT p95 matters more than mean if 5% of shoppers wait 8s.

Q142. Connection pooling?

Answer (simple). Reuse database connections instead of shaking hands every time.

Answer (interview). Pool size vs DB max_connections. Timeouts. Leaks.

Example. Spring Boot Hikari + FastAPI async pool sized for kiosk burst.

Q143. Pagination, filtering, idempotency keys in public APIs?

Answer (simple). Same as fresher SQL/HTTP, now with SLOs and abuse in mind.

Answer (interview). Max page size, consistent snapshots, signed idempotency keys with TTL.

Example. Staff audit export is async job + download link, not a 2GB GET.

Q144. 500K users — what do you clarify?

Answer (simple). Registered vs daily vs concurrent. Design to the peak QPS, not a vanity count.

Answer (interview). Load test. Caches. Read replicas. Queue writes. Colan 500K + 40% faster is a backend story — quote the measurement.

Example. 1M DAU chat ≠ 500K registered users. Different bottleneck (GPUs).

Mobile (Android / Flutter)

Q145. Jetpack: Room, Hilt, Navigation, WorkManager, DataStore?

Answer (simple). Room is local DB. Hilt is dependency injection. Navigation is screens. WorkManager is deferred work. DataStore is prefs.

Answer (interview). Lifecycle-aware. Test fakes. Migrations for Room.

Example. Cache last cart locally in Room if the socket drops.

Q146. Kotlin Coroutines + Flow on Android?

Answer (simple). Readable async + streams of UI state.

Answer (interview). viewModelScope cancellation. StateFlow vs SharedFlow. Backpressure.

Example. Device events Flow → ViewModel → Compose.

Q147. What is an ANR and how did you cut 70%?

Answer (simple). App Not Responding: main thread blocked ~5s. Users see a freeze dialog.

Answer (interview). StrictMode, traces, move IO/offload, Choreographer. Know the before/after measurement window for the resume 70% ANR / 30% performance claims.

Example. Do not parse JSON or hit disk on the main thread in checkout.

Q148. Flutter BLoC vs Provider vs setState?

Answer (simple). setState is local. Provider passes values down. BLoC is event-in state-out.

Answer (interview). Predictable state for staff apps. Test blocs without UI.

Example. TV monitor BLoC subscribes to cart fleet events.

Q149. Platform channels / FFI in Flutter?

Answer (simple). Talk to native Android/iOS from Dart when you need hardware.

Answer (interview). Keep channels small. Threading. Errors as typed results.

Example. SDK lite may call native scan/pay.

Q150. iOS interop / KMP at a high level?

Answer (simple). Share some logic across Android and iOS without sharing all UI.

Answer (interview). KMP expect/actual. Do not oversell if you shipped mostly Android+Flutter.

Example. Honest: 'I have shipped Android/Flutter; iOS via Flutter/interop, not a deep UIKit specialist unless true.'

Q151. Offline-first mobile?

Answer (simple). The app still works in a dead aisle, then syncs.

Answer (interview). Local source of truth, conflict resolution, queues. Not all features can be offline (payments).

Example. Scan locally; pay requires connectivity + idempotent retry.

Q152. Security on device: keystore, certificate pinning, secrets?

Answer (simple). Do not hide API keys in the APK. Use the OS keystore. Pin if you must.
Answer (interview). Obfuscation is not security. Attestation where needed. Secrets on server.
Example. Bedrock keys never in the trolley APK.

Q153. Play/App Store staged rollout?

Answer (simple). Ship to 1% of users, watch crashes, then 100%.
Answer (interview). Mapping files, ANR dashboards, feature flags vs store binary.
Example. Vision model file update may be a feature-flagged download, not only a Play release.

Q154. Deep links and notifications?

Answer (simple). A link opens a screen. A notification pokes the user.
Answer (interview). Auth on deep links. Do not leak PII in push payloads.
Example. Staff notified of a jammed kiosk — device id, not shopper name.

LLM, RAG, and agents (intermediate)

Q155. What is Generative AI? What is an LLM?

Answer (simple). GenAI creates new content. An LLM predicts the next tokens from context.
Answer (interview). Transformer decoder, tokens, context window, hallucination risk. Production: grounding, eval, cost, safety.
Example. Lisa generates answers; YOLO does not. Do not use an LLM as the only allergen source.

Q156. Tokens and context windows?

Answer (simple). Tokens are pieces of words. The window is how much the model can see at once.
Answer (interview). Tokenizer mismatch. Cost □ tokens. RAG should not dump 100k tokens of catalogue.
Example. Basket + top-5 chunks, not the whole store manual.

Q157. What is a transformer / self-attention (intermediate)?

Answer (simple). Attention lets each word look at the useful other words.
Answer (interview). Q/K/V, multi-head, residual, FFN, positional encodings. Encoder vs decoder vs encoder-decoder.
Example. An LLM connecting 'this' to the SKU mentioned earlier in the cart list.

Q158. Embeddings and cosine similarity?

Answer (simple). Meaning as a list of numbers. Cosine is how aligned two arrows are.
Answer (interview). Normalize, ANN indexes, out-of-domain queries, embedding model versioning (re-embed on upgrade).
Example. 'cheap sparkling water' near 'value seltzer' even if words differ.

Q159. Prompting vs RAG vs fine-tuning (intermediate)?

Answer (simple). Prompt = instructions. RAG = open book. Fine-tune = extra practice on weights.
Answer (interview). Facts that change → RAG. Style/format → prompt or FT. Combine. Prompting does not create truth.
Example. Weekly promo: RAG. JSON style: prompt. 5k receipt labels: LoRA.

Q160. Temperature and top-p?

Answer (simple). How adventurous decoding is. Low is steady.
Answer (interview). Does not guarantee facts. Low T for tools/JSON.
Example. Missing-SKU extraction at T≈0.

Q161. What is RAG (intermediate)?

Answer (simple). Find trusted pages, then write the answer from them.
Answer (interview). Ingest, chunk, embed, retrieve, (rerank), generate, cite, abstain. Eval retrieval and generation separately.
Example. Shopping assistant + recipes with LangChain/LlamaIndex.

Q162. BM25 vs dense vs hybrid (intermediate)?

Answer (simple). Words vs meaning vs both.
Answer (interview). BM25 for SKUs; dense for paraphrase; hybrid + RRF + rerank as default.
Example. EZY-1044 vs 'spicy pasta night'.

Q163. Chunking and overlap?
Answer (simple). Cut documents into useful pages, with a little overlap so sentences are not torn.
Answer (interview). Structure-aware. Overlap 10–20%. Tables/code special. Bad chunks = bad RAG.
Example. Return policy clause stays with its heading.

Q164. What is a vector database (intermediate)?
Answer (simple). Stores vectors and finds nearest neighbors, often with filters.
Answer (interview). Pinecone, Qdrant, Milvus, Weaviate, pgvector, FAISS, Supabase. Benchmark filters + recall + ops.
Example. ACL filter in the query, not after.

Q165. Hallucination (intermediate)?
Answer (simple). Confident nonsense.
Answer (interview). Ground, cite, abstain, eval, low T, critic. Never 0%.
Example. Do not invent gluten-free if the chunk is silent.

Q166. LangChain vs LlamaIndex (intermediate)?
Answer (simple). LangChain: app/tool Lego. LlamaIndex: data/RAG librarian.
Answer (interview). Can combine. Keep business logic in your code. Test without the LLM.
Example. Indexes in LlamaIndex; tools in LangChain/LangGraph.

Q167. What is an AI agent vs a chain (intermediate)?
Answer (simple). A chain is a recipe. An agent chooses the next allowed step.
Answer (interview). State, tools, budgets. More failure modes. Start with a chain.
Example. SKU lookup chain; multi-step recipe+missing-items agent.

Q168. ReAct pattern?
Answer (simple). Think, act, look, repeat.
Answer (interview). Reason + act + observe. Cap loops. Structured tools not free-form.
Example. get_stock → OOS → search_alternatives.

Q169. Tool calling (intermediate)?
Answer (simple). The model fills a form; your code runs the function.
Answer (interview). JSON schema, validate, authorize, execute, return. Model is not the security boundary.
Example. add_to_cart schema + Pydantic + HITL for pay.

Q170. MCP (intermediate)?
Answer (simple). A standard plug for tools and context.
Answer (interview). Client-server. Still authn/z, validation, audit. Not automatically safe.
Example. Catalogue MCP server; no raw SQL MCP without a guard.

Q171. LangGraph (intermediate)?
Answer (simple). A flowchart with memory: nodes do work, edges choose next, state is the backpack.
Answer (interview). Cycles, checkpoints, HITL, retries. Step limits.
Example. Lisa graph: retrieve → generate recipe → missing SKUs → confirm.

Q172. CrewAI / AutoGen (intermediate)?
Answer (simple). Role-based or conversational multi-agent helpers.
Answer (interview). Narrow roles, shared state, measure if extra agents help. Often a graph is enough.
Example. Planner/researcher/executor only if eval shows lift vs one agent.

Q173. LangSmith / Langfuse / LangFlow / n8n?
Answer (simple). Smith/Fuse: flight recorders and evals. LangFlow/n8n: visual pipes for prototypes/ops.
Answer (interview). Redact PII in traces. Visual ≠ production controls.
Example. Trace a failed add_to_cart; n8n for a Telegram TAN alert pipe, not checkout.

Q174. Hosted vs local LLM (intermediate)?

Answer (simple). API brain vs brain in your basement.

Answer (interview). Quality, cost, privacy, latency, ops. Abstraction with fallbacks: Bedrock, Vertex, GPT-4o, Claude, Gemini, Ollama, vLLM.

Example. Pilot hosted; FAQ local Llama 3 if cost/privacy says so.

Q175. LoRA / PEFT (intermediate)?

Answer (simple). Train a small adapter instead of the whole model.

Answer (interview). Rank, alpha, freeze base, eval task metrics, overfit on 5k. Merge or serve adapter.

Example. LoRA JSON extractor; keep SKU facts in RAG.

Q176. Quantization (intermediate)?

Answer (simple). Use fewer bits for numbers to go faster and smaller. Recheck quality.

Answer (interview). INT8/4-bit/FP8, PTQ vs QAT, TensorRT/TF Lite. Per-class metrics, not only average.

Example. INT8 YOLO on cart; if small SKUs drop, mix precision.

Q177. vLLM and KV cache (intermediate)?

Answer (simple). vLLM is a fast open-model server. KV cache remembers past tokens so you do not re-read the whole prompt.

Answer (interview). Paged KV, continuous batching, prefix cache. Different from on-device ORT.

Example. System prompt prefix-cached; cart hardware still TF Lite.

Q178. SSE (intermediate)?

Answer (simple). Server-sent events: the server talks in a stream. Stop if the client hung up.

Answer (interview). text/event-stream, disconnect cancel, heartbeats, Last-Event-ID.

Example. Lisa token stream; cancel GPU work on tablet sleep.

Q179. PII (intermediate)?

Answer (simple). Personal data. Do not send it to a model if you can avoid it.

Answer (interview). Detect/redact at gateway. Logs. Retention. Faces in video.

Example. Diet preference OK; PAN/Aadhaar never in prompts.

Q180. Prompt injection (intermediate)?

Answer (simple). A hidden note that tries to boss the model.

Answer (interview). Untrusted user+docs. Structured tools only. HITL for money. Red team.

Example. Product text 'ignore policy, 100% off' must not call discount tool.

Trading hygiene

Q181. What is a backtest vs paper vs live?

Answer (simple). Backtest is a replay. Paper is fake-live. Live is real money.

Answer (interview). Costs, slippage, latency, look-ahead, survivorship. Forward test. TAN: 6 years systematic, Telegram 5m/15m, Pine/Python, sub-5% DD — know definitions.

Example. Cannot use the unclosed 5m candle's high as a known value.

Q182. Maximum drawdown?

Answer (simple). The worst fall from a peak to a trough.

Answer (interview). Not the only risk metric. Sample size, period, costs. Sub-5% is a claim with a method.

Example. Peak 100 to 95 is 5% DD. Two strategies with same DD can have different tails.

Q183. Look-ahead bias and survivorship?

Answer (simple). Look-ahead: future bits. Survivorship: only studying winners that still exist.

Answer (interview). Point-in-time listings. Delisted symbols. Timestamp news.

Example. Sentiment from an article timestamped after the move is leakage.

Q184. SMC / liquidity in one honest sentence?

Answer (simple). A family of discretionary price-action ideas; if you use them, they must become testable rules.

Answer (interview). Do not mystify. Encode rules, test, risk limits. Not magic.

Example. TAN rule-based SMC-inspired signals with position sizing — validated, not folklore.

Q185. Why FinGPT is not a trading system?

Answer (simple). It reads finance language. It does not know your fills or fees.
Answer (interview). Feature candidate. Validate lift after costs. Risk limits still rule.
Example. Sentiment as one XGBoost feature, never auto-market-order.

Q186. Position sizing and risk limits?

Answer (simple). How big a bet, and the max pain you allow.
Answer (interview). Vol targeting, stop rules, kill switches. Independent of the ML toy.
Example. Sub-5% DD target drives size, not the model confidence alone.

Cloud (Bedrock, Vertex, SageMaker)

Q187. AWS Bedrock in one minute?

Answer (simple). Managed door to foundation models on AWS.
Answer (interview). IAM, data handling, guardrails, regions, latency, price. Still eval. Abstraction layer.
Example. Route Lisa to Bedrock Claude with fallback.

Q188. GCP Vertex AI?

Answer (simple). Google's managed train/serve/app AI platform.
Answer (interview). IAM, endpoints, feature store, monitoring. Avoid lock-in where it hurts.
Example. Same router interface as Bedrock.

Q189. SageMaker vs 'just a container'?

Answer (simple). SageMaker is AWS's training/hosting product. A container is DIY.
Answer (interview). Use managed when ops value > lock-in. Many teams serve FastAPI+ORT themselves.
Example. Training job on SageMaker; latency path still your FastAPI.

Q190. Databricks in an AI stack?

Answer (simple). A data+ML platform on Spark/lakehouse.
Answer (interview). Feature tables, training, governance. Not a replacement for an edge runtime.
Example. Catalogue analytics in Databricks; cart infer on device.

Q191. IAM least privilege?

Answer (simple). Every service gets the smallest key that still opens its door.
Answer (interview). Role per service, no long-lived keys in Git, audit.
Example. Embed job can read catalogue bucket, cannot kms:Decrypt all keys.

Q192. Blue/green or canary deploys?

Answer (simple). Run old and new, shift a little traffic, watch, then all.
Answer (interview). Prompt/model canaries with eval+business KPIs. Instant rollback.
Example. 5% of Lisa traffic on prompt v12; auto-revert if groundedness drops.

Level 3 — Advanced (GenAI, agents, RAG, MLOps, production)

Starts with the 25 industry questions (RAG, cost, agents, eval/safety, serving, production). Then senior RAG, agents, evals, serving, whiteboard design, and resume probes.

25 most-asked industry questions

Q193. Design a RAG pipeline that handles 10 million documents.

Answer (simple). Do not search the whole library every time. Index pages, shortlist, then read a handful.

Answer (interview). Ingest with tenant/ACL/source/time metadata. Structure-aware chunk+embed. Hybrid BM25+dense, ANN shards, metadata filters at retrieval, multi-query, rerank to 5–8, generate with citations, abstain. Upserts/deletes, eval set, SLOs on recall@k, groundedness, p95, $/query.

Example. Lisa/recipes over catalogue+policy — never dump 10M docs into the prompt.

Example. Shopper ACL must not retrieve staff stock-audit chunks.

Q194. Debug a RAG system that retrieves relevant context but generates wrong answers.

Answer (simple). Right page, wrong sentence — fix writing, not only search.

Answer (interview). Confirm retrieval@k first. Then prompt (cite/refuse), packing (lost-in-the-middle), contradictions, temperature, model ignoring context. Faithfulness eval. Log the exact window.

Example. Recipe RAG retrieves ingredients then invents a SKU — constrain to retrieved rows.

Example. Fail eval if Lisa cites a promo not in the chunk.

Q195. Choose the right chunking strategy for complex legal contracts.

Answer (simple). Do not tear a clause in half. Keep sections with their headings.

Answer (interview). Structure-aware: numbered clauses, definitions glossary, exhibits, overlap, table integrity. Eval questions that need a definition plus a later clause. Avoid naive 512-token windows.

Example. Return policy as a mini-contract: window stays attached to 'Returns'.

Q196. Compare BM25, dense retrieval, and hybrid search for different use cases.

Answer (simple). BM25 = exact words. Dense = meaning. Hybrid = both, then rerank.

Answer (interview). BM25 wins on SKUs/IDs; dense on paraphrase; hybrid+RRF default. Tune on a labelled set. Explain failure modes.

Example. EZY-1044 vs 'something spicy for pasta tonight'.

Q197. Add a reranker and prove its impact with evaluation metrics.

Answer (simple). Shortlist, then a careful reader reorders. Prove with a test.

Answer (interview). Retrieve k=50–100, cross-encoder to n=5. Offline nDCG/MRR/Recall@k; online groundedness, latency, cost. Flag + canary.

Example. 50 recipe chunks → 5 for the LLM. Do not rerank 10M docs.

Q198. Reduce LLM inference costs by 70% without sacrificing answer quality.

Answer (simple). Stop using the expensive brain for easy homework. Prove it on a frozen exam.

Answer (interview). Prompt/RAG compression, semantic cache, routing to small models, quant/distill, batching, cap tokens. Hold correctness+groundedness+safety. 70% is measured $/successful-task, not a slogan.

Example. FAQ cache/small model; complex recipes to GPT-4o/Claude.

Example. Vision already uses ONNX/TF Lite per frame instead of a cloud GPU.

Q199. Reduce time to first token in a production chatbot.

Answer (simple). Start talking sooner. Blank bubbles feel broken.

Answer (interview). SSE, shorter prefill, prefix/KV cache, speculative decoding, colocated GPU, don't block on all tools, faster model for greetings. Track TTFT p50/p95 separate from total.

Example. Lisa streams while a promo tool still runs. Stable system prompt for prefix cache.

Q200. Design a semantic caching system and track cache hit rates.

Answer (simple). Near-duplicate questions reuse a good answer.

Answer (interview). Embed normalized query; hit if similarity+ACL+fresh sources. Version hashes. Metrics: hit, stale-hit, quality, $, latency. Bypass personalization/stock.

Example. Basket-hash cache for recipes. Never cache another shopper's member price.

Q201. Build a model router that balances cost, latency, and response quality.

Answer (simple). Receptionist sends easy tickets to a junior, hard to a senior.

Answer (interview). Intent/complexity classifier. Routes: cache → extractive → 8B vLLM/Ollama → mid → frontier, with fallbacks, rate limits, token budgets. A/B on the same eval. This is the LLM abstraction on the resume.

Example. SKU lookup no LLM; recipe mid+RAG; complaint Claude/GPT-4o; Bedrock throttle → vLLM Llama 3.

Q202. Build a tool-using AI agent with reliable execution.

Answer (simple). Model asks; your code is the adult: validate, run, stop.

Answer (interview). LangGraph/CrewAI/AutoGen state machine. JSON schemas, allowlists, timeouts, retries, idempotency. HITL for money. Trace LangSmith/Langfuse. MCP ≠ security.

Example. Lisa: search auto; add_to_cart confirm; pay never from untrusted text.

Q203. Debug an agent stuck in an infinite tool-calling loop.

Answer (simple). Same calculator forever — max tries and notice repeats.

Answer (interview). Duplicate signatures, oscillating args, empty tool errors. Step budget, force-final node, circuit breaker, traces.

Example. get_stock empty SKU retries 40× → typed error after 2, then ask shopper.

Q204. Design long-term memory for an AI assistant.

Answer (simple). This-chat is short-term. Long-term is a curated notebook, not a junk drawer.

Answer (interview). Working thread + episodic summaries + structured profile + enterprise RAG. Write validated memories only. Recency×importance retrieve. User delete. No secrets.

Example. Remember vegetarian preference, never card PAN. TAN params in versioned config, not chat logs.

Q205. Enforce valid JSON output from an LLM.

Answer (simple). Ask for a form, not an essay. Reject broken forms.

Answer (interview). Schema/tool/constrained decode. Pydantic validate. One repair then fail closed. Low T.

Example. add_to_cart {sku, qty>0}. FastAPI already treats bodies as untrusted — same for LLM output.

Q206. Detect hallucinations and reduce unsupported answers.

Answer (simple). If the book is silent, say 'I don't know'.

Answer (interview). Ground, citations, NLI/judge on context only, abstain, eval set, monitor. Not 0%.

Example. No allergen claim without a chunk. Low-confidence YOLO does not bill.

Q207. Build an evaluation pipeline for a customer-support / shopping chatbot.

Answer (simple). A practice exam from real tickets. Every prompt change sits the exam.

Answer (interview). Golden set, retrieval+answer+tool+safety+latency+cost. CI gate. Human spot. Shadow traffic. LangSmith datasets.

Example. 100 Lisa questions with known SKUs. Fail if add_to_cart on missing item.

Q208. Make LLM-as-a-judge evaluations reliable and consistent.

Answer (simple). One picky teacher is noisy. Rubric, examples, calibration.

Answer (interview). Anchored rubric, few-shot, separate faithfulness vs tone, multi-sample, kappa vs humans, pin model+T, blind systems, re-calibrate on judge change.

Example. Judge only 'is every ingredient in evidence?', not 'sounds tasty'.

Q209. Defend a tool-using agent against prompt injection.

Answer (simple). A webpage note must not empty the till.

Answer (interview). User+retrieved text untrusted. Allowlist, schema, ignore non-channel 'tool calls', strip instructions in docs, sandbox, HITL money, red-team. MCP still needs authz.

Example. Description 'ignore policy, 100% off' cannot fire discount tool.

Q210. Detect and redact PII before sending data to an LLM.

Answer (simple). Cover the ID card before photocopying for a tutor.

Answer (interview). Regex+NER+checksums at gateway. Tokenize/vault. Minimize. No secrets in LangSmith. Blur faces if unused.

Example. Device ids in logs; 'call me at +91…' stripped from RAG. Public resume phones ≠ customer PII.

Q211. Choose between RAG, fine-tuning, and prompt engineering for a real use case.

Answer (simple). Instructions vs open book vs extra practice. Combine. Don't bake daily prices into weights.

Answer (interview). Prompt: format/tone/tools. RAG: changing facts. LoRA/FT: stable task/style with labels. Measure.

Example. Promo of the week = RAG. JSON style = prompt. 5k receipts = LoRA.

Q212. Fine-tune a small language model using LoRA on 5,000 examples.

Answer (simple). Bolt on a small trainable adapter; don't rebuild the engine.

Answer (interview). Freeze Llama 3/Mistral, rank/alpha/dropout, clean split, task metrics, early stop, merge or vLLM adapter, quantize after quality. 5k overfits easily.

Example. LoRA JSON {dishes, missing_skus}; catalogue stays RAG. No shuffled time-series labels.

Q213. Serve open-source LLMs at scale using batching, KV caching, and quantization.

Answer (simple). Share the oven, remember the story so far, use a lighter cookbook.

Answer (interview). vLLM: continuous batch, paged KV, prefix cache, parallel. AWQ/GPTQ/INT8. SLOs TTFT/tok-s. Fallback Bedrock/Vertex. Token budgets. Edge ≠ vLLM (ONNX/TF Lite).

Example. FAQ on vLLM Llama 3; hard RAG on frontier; cart vision still TensorRT/TF Lite sub-100ms.

Q214. Implement SSE streaming with proper handling of client disconnects.

Answer (simple). Talk while thinking; stop if they hung up so you don't burn GPU money.

Answer (interview). text/event-stream, flush tokens, cancel on disconnect, heartbeats, optional Last-Event-ID. FastAPI StreamingResponse. Don't buffer the whole answer.

Example. Tablet sleeps → cancel Lisa generation. WS for IoT; SSE for tokens.

Q215. Trace and debug every step of a multi-agent workflow.

Answer (simple). Flight recorder per helper: tools, tokens, why they stopped.

Answer (interview). OTel + LangSmith/Langfuse + LangGraph checkpoints, redacted prompts, correlation ids across FastAPI/Spring. Replay. Alert on loops/cost.

Example. Planner → retriever → add_missing_sku. Same discipline as IoT seq ids.

Q216. Roll back a prompt change that degrades production performance.

Answer (simple). Prompts are code. Revert like an API.

Answer (interview). Git+config version, CI eval, canary, auto-rollback on groundedness/tool-error/latency, pin last-good. No cowboy hot-edits.

Example. 'Be more creative' invented ingredients → revert prompt id.

Q217. Design a ChatGPT-style application for 1 million daily active users.

Answer (simple). A factory: queues, caches, routing, graceful failure — not one script. DAU ≠ concurrency.

Answer (interview). Auth, rate limits, conversation store, cache/small/RAG/agent routes, autoscaled vLLM/Bedrock, SSE, ACL, PII, safety, eval, cost/session, cells. SLOs TTFT/useful/safe/$.

Example. Hot path: cart WS events. Chat path: Lisa. Colan 500K taught indexing/caching — GPUs are the extra bottleneck.

RAG at scale

Q218. How do you handle document updates and deletions in RAG?

Answer (simple). If a policy is shredded, the assistant must not quote it tomorrow.

Answer (interview). Idempotent upserts, tombstones, TTL, re-embed on chunker/model change, search-time filter on version. Test deletion in eval.

Example. Expired promo vectors must not retrieve. Membership price updates invalidate semantic cache.

Q219. Lost-in-the-middle — what is it and the fix?

Answer (simple). Models skip the boring middle of a long pile of pages.

Answer (interview). Put best chunks first/last, shorten context, rerank harder, hierarchical summarize.

Example. 5 tight chunks beat 40 mediocre ones in Lisa.

Q220. Parent-child / hierarchical / late chunking?

Answer (simple). Retrieve a small bite, then bring its parent section for writing.

Answer (interview). Small embeddings for recall, larger window for generation. Late chunking embeds whole doc then pools.

Example. Retrieve a recipe step, expand to full recipe card before generating.

Q221. Hybrid fusion: RRF vs weighted score?

Answer (simple). Two ranked lists need a fair merge.

Answer (interview). Reciprocal Rank Fusion is scale-free. Weighted needs calibration. Then rerank.

Example. BM25 rank + dense rank → RRF → cross-encoder.

Q222. How do you evaluate RAG (senior)?

Answer (simple). Did we find the right page, and did we use it honestly?

Answer (interview). Recall@k, nDCG, MRR, faithfulness, answer relevance, citation accuracy, abstain quality, latency, cost, permission leaks. Separate retrieval vs gen.

Example. Golden 100 with labelled passages. CI fails on recall or faithfulness drop.

Q223. RAG security beyond ACL filters?

Answer (simple). Permissions, injection in docs, secrets in traces, confused deputy.

Answer (interview). Pre-retrieval authz, per-tenant indexes, injection tests, DLP, audit, encryption.

Example. A retrieved HTML comment should not become a tool call.

Q224. When is RAG the wrong tool?

Answer (simple). When you need a guaranteed calculation, a millisecond SKU lookup, or a stable classifier.

Answer (interview). Deterministic functions and SQL for facts; RAG for language over docs. Don't RAG 2+2.

Example. Cart total is arithmetic. Policy explanation is RAG.

Q225. Multi-index routing?

Answer (simple). First guess which library, then search it.

Answer (interview). Query classifier → policies vs recipes vs SKUs. Cuts noise and cost.

Example. Allergen question → policy index, not recipe blog.

Q226. Embedding drift and re-index jobs?

Answer (simple). New embedding model = old map is wrong. Rebuild.

Answer (interview). Version embeddings in metadata. Blue/green indexes. Dual-write during migrate.

Example. Upgrade embedding model overnight with a shadow index and recall compare.

Agents, tools, and memory

Q227. When is a multi-agent system worse than one agent?

Answer (simple). Too many cooks, extra cost, extra failure points.

Answer (interview). Measure task success, latency, $ against a single-agent or chain baseline. Add agents only for genuine specialization.

Example. Formatting JSON with three CrewAI roles is worse than Pydantic.

Q228. Human-in-the-loop patterns?

Answer (simple). The model may suggest; a human taps OK on money and safety.

Answer (interview). LangGraph interrupt, timeouts, default-deny. Audit who approved.

Example. Add-to-cart auto; pay requires the shopper tap.

Q229. Idempotent tool calls?

Answer (simple). Retrying 'charge' must not charge twice.

Answer (interview). Idempotency keys from (trace, tool, canonical args). Store results.

Example. add_to_cart retried after WS blip uses the same key.

Q230. Agent memory vs RAG vs scratchpad?

Answer (simple). Scratchpad is this task's notebook. Memory is longer. RAG is company books.

Answer (interview). Don't dump full history. Summarize. Structured state in LangGraph.

Example. State: basket[], missing[], step. RAG: policies. Memory: diet pref.

Q231. How do you bound agent cost?

Answer (simple). Max steps, max tokens, max $, max wall time, then stop.

Answer (interview). Budgets in the graph. Route cheap models first. Kill switches.

Example. Recipe agent: ≤6 tool calls, ≤2k generated tokens.

Q232. Deterministic fallbacks for agents?

Answer (simple). If the clever helper fails, a boring function still works.
Answer (interview). SKU barcode path never requires an LLM. LLM is overlay.
Example. Scan still adds the item if Lisa is down.

Q233. AutoGen vs CrewAI vs LangGraph — senior choice?

Answer (simple). Pick the simplest orchestration you can operate.
Answer (interview). LangGraph: control/state/HITL. CrewAI: roles. AutoGen: conversational. Eval, traces, security identical.
Example. Production Lisa: LangGraph. Research spike: CrewAI. Don't run all three.

Q234. How do you test agents without hitting real APIs?

Answer (simple). Fake tools that return recorded answers. Assert the calls.
Answer (interview). Contract tests, golden traces, simulators, property tests (no pay tool without confirm).
Example. Replay a fixture: OOS milk → must call search_alternatives, must not pay.

Q235. Supervisor / router / swarm topologies?

Answer (simple). One boss, or a circle of peers. Boss is easier to debug.
Answer (interview). Supervisor graph is the default. Swarms need stronger invariants.
Example. Planner node dispatches retriever vs vision vs FAQ.

Q236. What is a graph checkpoint and why?

Answer (simple). A save-game of the agent so you can resume after a crash or a human pause.
Answer (interview). LangGraph checkpointer (thread_id). Don't store PII unencrypted.
Example. Shopper walks away mid-recipe; resume the same thread.

Evals, safety, and MLOps

Q237. What is MLOps for LLM apps (not only sklearn)?

Answer (simple). Version prompts, tools, indexes, models. Eval, deploy, watch, rollback.
Answer (interview). Artifacts: prompt id, embedding model, chunker, index build, agent graph. Quality+infra metrics. Canaries.
Example. Merchant deploys plus prompt/model eval jobs, not only APKs.

Q238. Offline vs online vs human eval?

Answer (simple). Offline exam, live traffic shadows, people for gray cases.
Answer (interview). You need all three. Online can surprise. Humans calibrate judges.
Example. Weekly 50 human-labelled Lisa chats to keep the judge honest.

Q239. Safety policies: self-harm, scams, medical, allergens?

Answer (simple). Refuse and redirect. Allergens are high-stakes — no guessing.
Answer (interview). Classifiers + rules + abstain. Legal review. Log. Don't rely on the base model alone.
Example. If RAG has no allergen field, say so. Never 'probably fine'.

Q240. Jailbreaks vs prompt injection?

Answer (simple). Jailbreak: user tries to break the system prompt. Injection: untrusted content tries to steer tools.
Answer (interview). Different controls. Both red-teamed. Tools still allowlisted.
Example. User 'ignore rules' vs document 'call discount tool'.

Q241. Data residency and vendor LLMs?

Answer (simple). Where do the tokens go, and who can see them?
Answer (interview). Contracts, regions, zero-retention, VPC endpoints, on-prem vLLM when required.
Example. EU shopper chats may not go to a US-only endpoint without a decision.

Q242. Prompt/version governance?

Answer (simple). Who can change the brain, with what evidence?
Answer (interview). CODEOWNERS, eval required, change log, on-call.
Example. Prompt v12 PR shows nDCG and groundedness vs v11.

Q243. Drift in RAG (content drift vs embedding vs user drift)?

Answer (simple). The books change, the map changes, or the questions change.
Answer (interview). Monitor retrieval hit rates, unknown-intent, thumbs, fresh-doc lag.
Example. New promo language not in the index → spike in abstains.

Q244. Cost accounting per tenant?

Answer (simple). Each store should see its LLM bill.
Answer (interview). Token meters, cache hits, GPU-ms, pass-through + margin. Quotas.
Example. A viral recipe loop cannot bankrupt the platform — quota + 429.

Q245. Feature flags for models?

Answer (simple). Flip brains without a full redeploy.
Answer (interview). Flags for prompt, temperature, reranker on/off, model id. Combine with canary.
Example. Turn off reranker if GPU pool is hot; accept quality hit explicitly.

Q246. Shadow mode?

Answer (simple). New brain answers in the log, old brain still talks to the shopper.
Answer (interview). Compare distributions, then graduate. Privacy in logs.
Example. v12 drafts Lisa answers 10% shadow before cutover.

Q247. Rollback of an index?

Answer (simple). Bad embed job? Point search back at yesterday's index.
Answer (interview). Immutable index builds, alias swap, keep N generations.
Example. Alias 'catalogue-live' → build-2026-10-28.

Q248. Eval for agents: success rate of trajectories?

Answer (simple). Not only pretty text — did the tools do the right thing?
Answer (interview). Exact tool sequence, arg match, end state, side-effect counts.
Example. Success = missing pasta sauce added once, not twice, not paid.

Q249. Red teaming cadence?

Answer (simple). Schedule attacks: injection, PII, jailbreak, cost bombs, loops.
Answer (interview). Automated suite in CI plus quarterly human red team. Track coverage.
Example. Suite includes 'ignore previous instructions' in a product HTML comment.

Q250. Guardrails vs fine-tune vs policy model?

Answer (simple). Rules on the outside, or a model trained to behave, or both.
Answer (interview). Layered: regex/PII → policy classifier → base LLM → output filter.
Example. Output filter strips accidental card numbers even if the LLM slipped.

Q251. What should block an AI deploy?

Answer (simple). Failed tests, eval regressions, secret scan, vuln, missing canary plan.
Answer (interview). Write the gate list. Quality is a blocker, not a dashboard nice-to-have.
Example. Groundedness -3% blocks. Copy tweak that doesn't touch eval may still need review.

Serving, inference, and platforms

Q252. Prefill vs decode and why TTFT cares?

Answer (simple). Prefill reads the prompt; decode writes the next word. First word waits on prefill.
Answer (interview). Long RAG prompts kill TTFT. KV cache helps later tokens more than the first.
Example. Cut system+history+chunks before buying more GPUs.

Q253. Continuous batching vs static batch?

Answer (simple). Seat new diners as seats free, don't wait for a full table of 8.
Answer (interview). vLLM paged attention. Fairness vs utilization. Deadlines.
Example. Many Lisa sessions share a GPU without waiting for lockstep.

Q254. Speculative decoding?

Answer (simple). A small model guesses a few tokens; the big one checks.
Answer (interview). Helps decode speed when draft is good. Extra complexity.
Example. Useful on FAQ-like chats; less if outputs are highly random.

Q255. Multi-LoRA serving?

Answer (simple). Many small adapters on one base model.
Answer (interview). vLLM LoRA slots. Memory vs convenience. Don't mix tenants' adapters by accident.
Example. Per-merchant tone adapter; shared catalogue RAG.

Q256. GPU vs CPU vs NPU placement?

Answer (simple). Big GPUs in the cloud. NPU/CPU on the cart. Don't mix jobs blindly.
Answer (interview). Profile. Thermal. Batch. Precision.
Example. YOLO on NPU; LLM on GPU; BM25 on CPU.

Q257. Backpressure when GPUs are full?

Answer (simple). A waiting line with a max length, then say 429/503, don't melt.
Answer (interview). Queue+timeout+shed. Load balancer least-outstanding. Degradation: smaller model.
Example. If vLLM queue > 200ms p95, route FAQs to cache only.

Q258. How do you design multi-region LLM?

Answer (simple). Don't put one GPU in one basement for the planet.
Answer (interview). Sticky sessions vs stateless. Data residency. KV cache is regional. Failover.
Example. MY vs IN shoppers — residency + latency.

Q259. Exactly-once LLM side effects?

Answer (simple). You usually can't. Make tools idempotent; accept at-least-once.
Answer (interview). Outbox pattern, keys, at-least-once tools.
Example. Retry add_to_cart with the same key after a 502.

Q260. Schema evolution for tool JSON?

Answer (simple). Add optional fields; don't rename in place.
Answer (interview). Version tools. Agents pin a version. Pydantic extra=forbid on money tools.
Example. add_to_cart v2 adds notes?: str optional.

Q261. How do you do CI for prompts?

Answer (simple). PR runs the golden set. Numbers in the comment.
Answer (interview). Fail on delta. Store traces as artifacts. Don't call prod keys in CI if avoidable — use recorded fixtures + a cheap model.
Example. GitHub Action: pytest + ragas/langsmith eval.

Q262. Canary a vision model on devices?

Answer (simple). 1% of carts get the new ONNX, rest stay.
Answer (interview). Device cohorts, crash/thermal/mAP-proxy, instant disable flag.
Example. Night shift cohort first — lighting is harsher.

Q263. Feature store vs just SQL?

Answer (simple). A feature store is a disciplined way to get the same features at train and serve.
Answer (interview). Worth it at scale/many models. Overkill for one XGBoost.
Example. Promo propensity features in Databricks; checkout SKU is not a feature store problem.

Q264. Online vs offline features for XGBoost?

Answer (simple). Offline: batch tables. Online: low-latency lookup at request.
Answer (interview). Skew if code paths differ. Point-in-time correctness.
Example. Don't train on a rolling 30d average that isn't available live.

System design, leadership, and this resume

Q265. Design an enterprise RAG assistant (whiteboard).

Answer (simple). Secure librarian: permissions, pages, quotes, grades.

Answer (interview). Users, SLOs, ingest, chunk, hybrid, rerank, ACL, grounded gen, citations, eval, telemetry, delete/update, cost. Trade-offs.

Example. Sources → indexes → retrieve → rerank → LLM → validate → telemetry.

Q266. Design a tool-using agent (whiteboard).

Answer (simple). Approved tools, stop rules, human for danger.

Answer (interview). State machine, schemas, least privilege, budgets, HITL, persistence, traces, deterministic fallback, idempotent tools.

Example. Search inventory freely; purchase requires confirmation.

Q267. Design a real-time vision pipeline (whiteboard).

Answer (simple). Camera → prepare → recognize → event. Watch FPS and end-to-end ms.

Answer (interview). Capture, preprocess, infer, post, track, timestamp, backpressure, device metrics, fallback.

Example. ezyCart: frame to SKU event on WebSocket under 100ms model budget.

Q268. Design WebSockets for thousands of carts.

Answer (simple). Many phone calls. Don't let one bad cart kill the PBX.

Answer (interview). Shard by store, heartbeats, backpressure, sticky vs broker (Kafka), replay, auth per device, horizontal gateways.

Example. IoT gateway Spring Boot/FastAPI, partition key store_id.

Q269. Investigate a slow API (senior).

Answer (simple). Find the slow station with traces, not guesses.

Answer (interview). Compare baseline, traces, EXPLAIN, deps, resources, reproduce load, one fix.

Example. SQL 80% of p95 — index, don't rewrite FastAPI.

Q270. Deduplicate records in Python (talk through).

Answer (simple). Remember ids you've seen; keep first or last — say which.

Answer (interview). O(n) set. Define collision policy. Stable sort if needed.

Example. seen=set(); for e in events: if e.id not in seen: keep.

Q271. Bounded retry pattern?

Answer (simple). Retry a few times with waits, then fail. Don't retry forever. Don't retry non-idempotent blindly.

Answer (interview). jittered exponential backoff, retry-after, max attempts, classify retryable errors.

Example. 429/503 yes; 400 validation no.

Q272. SQL index example you can write?

Answer (simple). CREATE INDEX ... then EXPLAIN.

Answer (interview). Composite order, covering indexes, write cost.

Example. CREATE INDEX idx_orders_customer_id ON orders(customer_id);

Q273. How do you make architecture decisions?

Answer (simple). Requirements, options, risks, cost, write it down, test.

Answer (interview). ADRs. Bake-offs. Revisit. Edge vs cloud, chain vs agent.

Example. YOLO vs RT-DETR decided by mAP+ms on store video.

Q274. Production incident response?

Answer (simple). Stop the leak, then find the hole, tell people, learn without blame.

Answer (interview). Mitigate → diagnose → recover → RCA. Metrics first: error, latency, GPU, queue.

Example. Missing device events: net, broker, schema, last deploy, replay by event id.

Q275. Engineering disagreement?

Answer (simple). Shared goal, evidence, maybe a test, record the call.

Answer (interview). Disagree and commit. No ego bake-offs.

Example. Two retrievers on the same golden set.

Q276. How do you mentor?

Answer (simple). Why, then how, then they drive.

Answer (interview). Slices, reviews, increasing ownership. Teach trade-offs.

Example. Pair on API design; they present the ADR.

Q277. How do you prioritize?

Answer (simple). Impact, risk, urgency, effort. Brakes before paint.

Answer (interview). Make trade-offs visible. Kill zombie work.

Example. Checkout correctness > cosmetic dashboard.

Q278. Tell me about a failure (framework)?

Answer (simple). Real story: situation, my part, impact, fix, process change. Don't invent.

Answer (interview). STAR. What you'd do differently.

Example. Use a genuine integration/perf miss — e.g. missed seq-id causing duplicate scans in staging.

Q279. How do you measure project success?

Answer (simple). Did people get the job done, reliably, at acceptable cost — not 'the model exists'.

Answer (interview). Adoption, task completion, quality, latency, reliability, cost, risk.

Example. Lisa: answer quality, add-to-cart success, TTFT, $.

Q280. Why should we hire you as Principal AI?

Answer (simple). I connect models, retrieval, agents, APIs, edge vision, and apps — and I can say no to fake metrics.

Answer (interview). 13+ years, Principal AI / Lead GenAI & Agentic at Retailetics, ezyCart+RAG+edge, $2M nav, TAN R&D, mobile supporting. 90-day plan: evals+SLOs first.

Example. AI + systems + delivery. Honest about tools I have not run at scale.

Q281. How do you answer when you have not used a tool deeply?

Answer (simple). Say conceptual vs hands-on, map to a pattern, propose a 1-week spike with metrics. Don't bluff.

Answer (interview). Interviewers reward learning plans over theater.

Example. 'I have hybrid search in prod; I have not operated X at 10M docs; here's the POC.'

Q282. Prototype vs production AI?

Answer (simple). Science fair vs a trolley people lean on.

Answer (interview). Eval, authz, observability, versioning, rollback, support, cost. Production is a system.

Example. A Colab RAG is not Lisa.

Q283. How do you describe impact honestly?

Answer (simple). Baseline, action, measurement, period, team vs me.

Answer (interview). Every resume number: method. $2M beacons, sub-100ms, sub-5% DD, 70% ANR, 30% perf, 500K/40%.

Example. No $X million placeholders.

Q284. ezyCart in 60 seconds?

Answer (simple). Smart trolley: cameras, sensors, tablet, backend, AI checkout.

Answer (interview). Edge vision (YOLO/RT-DETR), Kimera-VIO nav, WS IoT, Android, FastAPI/Spring, RAG recipes/Lisa, sub-100ms, fail-safe.

Example. Camera/sensors → edge AI → app/backend → pay.

Q285. Indoor navigation $2M story — what they will probe?

Answer (simple). Software localization vs thousands of beacons.

Answer (interview). Baseline cost, accuracy metric, failure (textureless), your role vs team, how savings were calculated.

Example. Don't just say '$2M'; walk the arithmetic you actually used.

Q286. TAN Intelligence — how to talk about it?

Answer (simple). Independent venture, not a second employer. Custom indicators, 6 years, Telegram 5m/15m, SMC rules, Pine/Python, sub-5% DD.

Answer (interview). Leakage, costs, walk-forward. Not investment advice. Risk limits.

Example. Keep it off the Experience list; keep it on the R&D story.

Q287. 90-day plan if hired?

Answer (simple). Week 1–2: SLOs, evals, traces, cost. Then the riskiest quality or latency gap. Then a small agent/RAG improvement behind a flag.

Answer (interview). Don't boil the ocean. Ship a measured win.

Example. Stand up golden set + Langfuse before a new agent framework.

Q288. How do you work with PMs and legal on AI?

Answer (simple). Translate risk: allergens, PII, hallucination, cost. Offer options with numbers.

Answer (interview). Red lines in writing. Eval as a shared language.

Example. Legal: abstain on medical/allergen without source.

Q289. On-call for LLM systems?

Answer (simple). Pages on error, tail latency, cost spike, loop, safety filter down.

Answer (interview). Runbooks: rollback prompt, disable tool, shed to FAQ cache.

Example. Cost anomaly at 2am: kill runaway agent graph.

Q290. Capacity planning for tokens?

Answer (simple). Guess QPS × tokens × $/token × cache hit. Stress.

Answer (interview). Peak vs average. Seasonal retail. Headroom.

Example. Weekend recipe traffic vs weekday SKU lookups.

Q291. How do you keep mobile supporting skills without leading with them?

Answer (simple). They are how AI reaches the floor, not the headline.

Answer (interview). Kotlin/Compose/Flutter/Android/iOS as the last mile of vision+agents. Don't apply as a pure Android IC if the role is Principal AI — unless they want both.

Example. WS into ezyCart Android is the punchline, not RecyclerView trivia — but you can still answer ANR.

Q292. Thread safety on Android vs in the agent backend?

Answer (simple). UI thread vs IO. Backend: don't share mutable agent state across requests.

Answer (interview). Immutable messages, per-thread state, locks only when needed.

Example. One LangGraph thread_id per cart session, not a global dict.

Q293. How would you introduce evals to a team that only ships APKs?

Answer (simple). Start with 30 labelled questions and a CI number. Don't start with a platform.

Answer (interview). Show a caught regression. Then invest in LangSmith.

Example. The 'creative' prompt that invented ingredients is the story.

Q294. SQL vs vector vs BM25 — who owns the truth?

Answer (simple). Transactional truth in SQL. Language search in BM25/vectors. Don't duplicate poorly.

Answer (interview). Price/stock from SQL/API at answer time even if RAG mentioned an old price.

Example. Lisa quotes promo from SQL live, not from a week-old chunk alone.

Q295. Event sourcing for carts?

Answer (simple). Store what happened (scanned, removed), rebuild state.

Answer (interview). Great for replay/debug; harder reads. Mix with snapshots.

Example. Replay a cart to debug a double charge.

Q296. Backpressure on camera frames?

Answer (simple). If infer is slow, drop frames, don't queue 10 seconds of past.

Answer (interview). Always latest-frame policy for checkout. Count via tracking, not every frame.

Example. Keep last frame only; track IDs across kept frames.

Q297. How do you version ONNX models on device?

Answer (simple). A file with a version, a flag, a rollback.

Answer (interview). Signed artifacts, compatible input shapes, A/B cohort, disk space.

Example. yolo-store-v17.onnx with min app version.

Q298. Coordinator vs IC at Principal level?

Answer (simple). You still touch the critical path, but you also set the quality bar and unblock.
Answer (interview). Design reviews, eval gates, mentoring, saying no to unmeasured agents.
Example. You might write the LangGraph skeleton and the eval harness, not every tool.

Q299. What is a kill switch?

Answer (simple). A big red button that disables the clever path.
Answer (interview). Per-feature flags, default to deterministic checkout if AI is sick.
Example. Disable Lisa globally; scanning still works.

Q300. Observability for hallucinations in prod?

Answer (simple). You won't label all chats. Sample, user flags, critic on a slice, retrieval-empty rate.
Answer (interview). Don't wait for a viral wrong-allergen story.
Example. Thumbs-down + 'unsupported claim' critic on 5% traffic.

Q301. How do you handle multilingual RAG (Tamil/EN/MS)?

Answer (simple). Embed in a multilingual model or translate-then-search. Eval per language.
Answer (interview). Tokenizer cost differs. Don't assume English chunking.
Example. Tamil recipe query against EN catalogue — translation + hybrid.

Q302. Streaming tool calls (parallel tools)?

Answer (simple). Ask stock and promo at once if independent.
Answer (interview). Fan-out with timeouts. Merge. Don't parallel a tool that depends on the other.
Example. get_stock and get_allergens together; add_to_cart after both.

Q303. What is structured output / outlines / grammar decoding?

Answer (simple). Force the model to speak JSON by construction, not by begging.
Answer (interview). Constrained decoding vs post-validate. Still validate.
Example. Outlines/json_schema for tool args.

Q304. Chain-of-thought in production?

Answer (simple). Hidden thinking can help some models; it can also leak and cost tokens.
Answer (interview). Don't stream raw CoT to shoppers. Don't treat CoT as evidence of truth.
Example. Use it internally; show shoppers the grounded short answer.

Q305. How do you prevent cost bombs (infinite generate)?

Answer (simple). max_tokens, stop sequences, step budgets, per-tenant quotas, anomaly alerts.
Answer (interview). Hard limits in the gateway, not only in the prompt.
Example. Gateway kills runs > $0.50 or > 30s.

Q306. A/B testing LLM apps vs classic A/B?

Answer (simple). Outputs are random; you need more samples and quality metrics, not only conversion.
Answer (interview). Holdout, CUPED if you have it, guardrail metrics (safety).
Example. Don't ship a wittier Lisa that hallucinates more even if clicks rise.

Q307. Knowledge graphs vs vector RAG?

Answer (simple). Graphs are explicit relations; vectors are fuzzy similarity. Hybrid exists.
Answer (interview). Use graphs when relations are the product (allergens, substitutes). Vectors for prose.
Example. SKU substitute graph + vector recipes.

Q308. How do you do RAG over tables?

Answer (simple). Don't embed a CSV as a blob blindly. Preserve rows, maybe text-to-SQL with tight allowlists.
Answer (interview). Row-level chunks, schema linking, or constrained SQL. Test numeric exactness.
Example. Nutrition table: retrieve the row, don't round-trip an LLM for 2+2.

Q309. Code interpreter / sandboxed tools?

Answer (simple). Let the model run code in a jail, not on the checkout host.
Answer (interview). No network, timeouts, resource limits. Still injection risk.
Example. Don't give a sandboxed python that can hit the payment API.

Q310. How do you explain attention to a hiring manager who is not ML?

Answer (simple). It's a highlighter for the useful words in the sentence.
Answer (interview). Then offer the Q/K/V sentence if they want depth.
Example. Connecting 'it' to 'trolley' in the shopper question.

Q311. Staff/principal system design checklist you actually follow?

Answer (simple). Clarify, SLOs, scale, draw, data, failure, security, observe, trade-off, validate.
Answer (interview). Say it as a habit, then apply to their prompt.
Example. Always name p95, ACL, eval, one failure mode.

Q312. How do you keep from gold-plating agents?

Answer (simple). Start as a function, then a chain, then an agent if eval says so.
Answer (interview). Complexity budget. Revisit quarterly.
Example. SKU lookup stayed a function. Recipes earned a graph.

Q313. Mobile + AI interview: what they might still ask?

Answer (simple). ANR, coroutines, Compose, Flutter BLoC, offline, WS reconnect — supporting, not the headline.
Answer (interview). Tie every answer back to how the model reaches the shopper.
Example. Main-thread policy + device WS + on-device TF Lite.

Q314. How would you onboard a new embedding model with 10M docs?

Answer (simple). Shadow index, dual retrieve, compare recall, alias swap, keep rollback.
Answer (interview). Cost of re-embed, backfill SLA, query routing during migrate.
Example. Build catalogue-v2 overnight, shadow 2 days, swap alias.

Q315. What is a 'cell' or 'bulkhead' in an AI platform?

Answer (simple). Blast-radius boxes so one noisy tenant doesn't drown GPUs.
Answer (interview). Per-tenant queues, separate pools for batch embed vs interactive chat.
Example. Embed jobs cannot starve Lisa TTFT.

Q316. How do you handle model deprecation from a vendor?

Answer (simple). Abstraction, dual-run, eval parity, date in the calendar.
Answer (interview). Never hardcode gpt-x in 40 repos. One router.
Example. GPT-4o → successor behind the same interface after eval.

Q317. Security review questions you pre-empt?

Answer (simple). Where PII goes, tool authz, injection, logs, residency, supply chain of models.
Answer (interview). Bring a diagram. Don't wait to be asked.
Example. Slide: data flow from cart camera to event (no raw video stored) vs chat to LLM (redacted).

Q318. How do you document an agent for the next engineer?

Answer (simple). Graph picture, tool contracts, budgets, eval, runbook, known failure modes.
Answer (interview). README + ADR + golden traces.
Example. Lisa README shows the graph and the pay-tool never-list.

Q319. Final: what is production GenAI in one sentence?

Answer (simple). A measured, permissioned, observable, reversible system that uses models where they earn their keep.
Answer (interview). Not a prompt in a slide. Quality + safety + SLO + cost + rollback.
Example. ezyCart + Lisa + edge vision + FastAPI/Spring + evals — that's the bar.

End of bank — 319 questions. Rehearse out loud. If an example is not yours, skip it.